# Raihan Anwar

## Project Automation and Improvement Assistant Section Head

Industrial Automation & IIoT Engineer with 5+ years of experience in manufacturing industry, specializing in OEE digitalization, smart factory implementation, and machine-to-database integration. Proven track record in increasing production efficiency, leading cross-functional teams, and implementing Industry 4.0 initiatives. Strong background in PLC, SQL, Python, and data-driven manufacturing improvement.


anwarraihan10@gmail.com

(+62) 85766600726

Tangerang, Indonesia

28, November 1996

linkedin.com/in/RaihanAnwar


## WORKING EXPERIENCE
## PT PANATA JAYA MANDIRI

### Process Engineering
Sub Section Head

*03/2020 - 12/2022* *Tangerang*

*Achievements/Tasks*

- Implemented Lean Manufacturing tools (Kaizen & 5 Why analysis)
- Ensured projects were delivered on schedule and within target objectives
- Reduced material waste by 8% through process optimization and root cause analysis

### Sales And Marketing Local Customer
Assistant Section Head

*01/2022 - 02/2023* *Tangerang*

*Achievements/Tasks*

- Coordinated production and logistics teams to ensure 100% on-time delivery performance
- Acted as liaison between customers and internal departments to align technical and commercial requirements
- New Product Development with customer requested

### Project Automation and Improvement
Assistant Section Head

*01/2023 - Present* *Tangerang*

*Achievements/Tasks*

- Increased production line OEE by 12% through real-time machine monitoring system
- Developed machine-to-database integration using PLC, SQL, and Node-RED
- Implemented IIoT architecture for production tracking and downtime analysis
- Leading digital transformation initiative aligned with Industry 4.0 roadmap

## SKILLS

Lean Manufacturing 4.0 | IIoT | PLC | SQL | Node Red | Visual Studio | Python | Java | Data Engineer | Electrical Wiring And Installation | Hydraulic & Pneumatic | Automation | Microsoft Office | Team Management | Project Management | Problem Solving

## EDUCATION

### Politeknik Negeri Sriwijaya
Mechatronic Engineering

*09/2014 - 09/2018* *GPA 3.35*

## CERTIFICATE

Lean MONOZUKURI 4.0 (Kemenperin and AOTS Certificate) (09/2023 - 09/2028)
*PIDI 4.0 Kemenperin*

Training Pneumatic and Hydraulic Batch 2 (BNSP Certificate) (06/2023 - 06/2028)
*PIDI 4.0 Kemenperin*

Training Cloud Engineer (BNSP Certificate) (04/2024 - 04/2029)
*PIDI 4.0 Kemenperin*

Data Scientist (BNSP Certificate) (10/2025 - 10/2030)
*PIDI 4.0 Kemenperin*

## HONOR AWARDS

2nd Championship ADROPEN (08/2024)
*ADRGroup of Companies*

2nd Championship ADROPEN (08/2025)
*ADRGroup of Companies*

## LANGUAGES

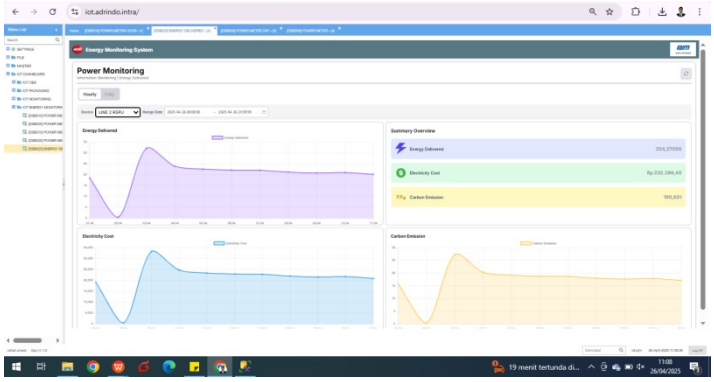
English
*Professional Working Proficiency*

Indonesia
*Professional Working Proficiency*

# ELECTRICITY MONITORING DASHBOARD

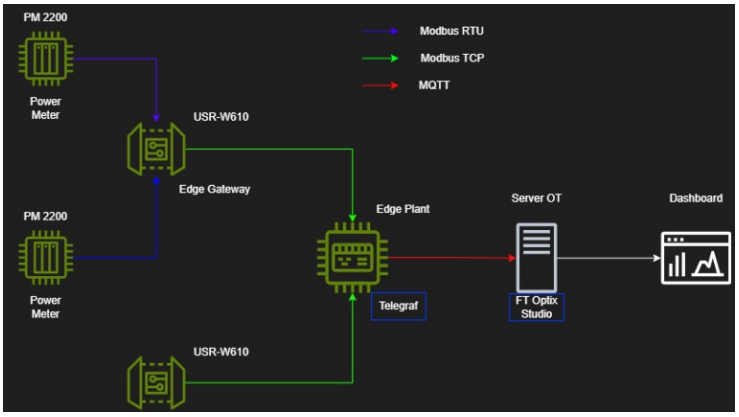
**Project Overview**

Developed an electricity monitoring dashboard for one production plant with 30 critical high-consumption machines. The system enables real-time monitoring of energy usage and supports data-driven energy efficiency improvement.




Dashboard page summary

The electricity monitoring system was designed using Schneider PM2200 power meters installed across 30 critical high-energy consumption machines within the production plant. Energy data collected via Modbus RTU was converted to Modbus TCP to improve communication reliability and simplify network integration. The devices were connected to an industrial PC acting as an edge device within the plant network. An ETL process was implemented using Telegraf to acquire, process, and forward the data into both SQL Database and Time-Series Database (TSDB) for structured storage and historical analysis. The processed data was then transmitted to a central server where it was integrated into a dashboard visualization system and data warehouse environment, enabling real-time monitoring, historical trend analysis, and energy performance evaluation.




Architecture Network and Device

PM 2220 in Panel Machine

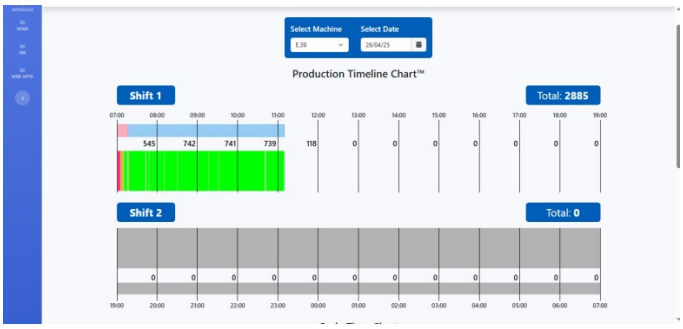
USR-W610 In Panel Distribution

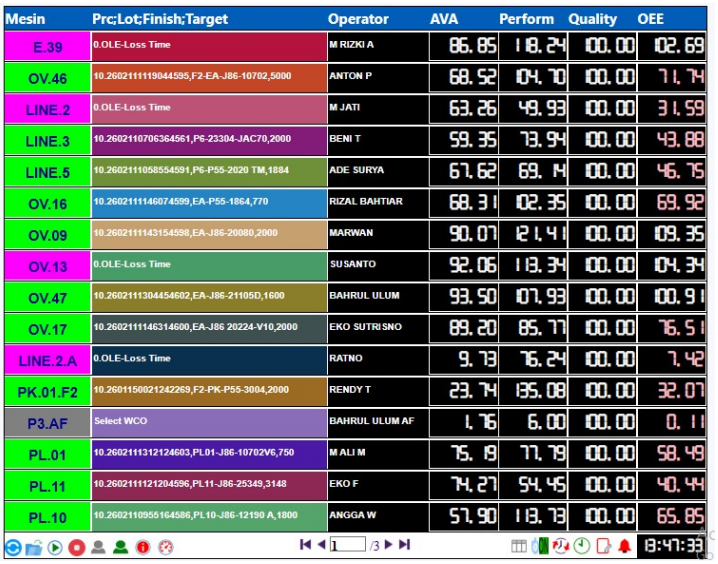
PC Industrial In Production Plant Office

# OEE MONITORING DASHBOARD

**Project Overview**

Designed and implemented real-time OEE monitoring system for 52 production machines. The system enables management to monitor downtime history, production quantity, cycle time, and machine speed.

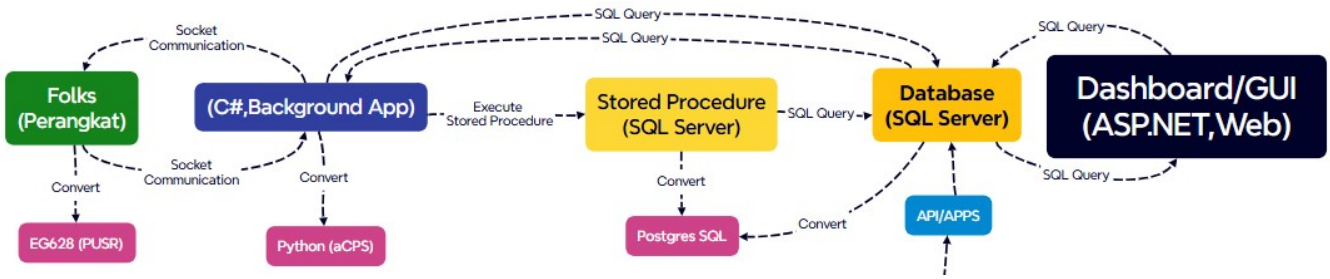




| Mesin | Prc;Lot;Finish;Target | Operator | AVA | Perform | Quality | OEE |
|---|---|---|---|---|---|---|
| E.39 | 0.OLE-Loss Time | M RIZKI A | 86.85 | 118.24 | 100.00 | 102.69 |
| OV.46 | 10.2602111119044595,F2-EA-J86-10702,5000 | ANTON P | 68.52 | 104.70 | 100.00 | 71.74 |
| LINE.2 | 0.OLE-Loss Time | M JATI | 63.26 | 49.93 | 100.00 | 31.59 |
| LINE.3 | 10.2602110706364561,P6-23304-JAC70,2000 | BENI T | 59.35 | 73.94 | 100.00 | 43.88 |
| LINE.5 | 10.2602111058554591,P6-P55-2020 TM,1884 | ADE SURYA | 67.62 | 69.14 | 100.00 | 46.75 |
| OV.16 | 10.2602111146074599,EA-P55-1864,770 | RIZAL BAHTIAR | 68.31 | 102.35 | 100.00 | 69.92 |
| OV.09 | 10.2602111143154598,EA-J86-20080,2000 | MARWAN | 90.07 | 121.41 | 100.00 | 109.35 |
| OV.13 | 0.OLE-Loss Time | SUSANTO | 92.06 | 113.34 | 100.00 | 104.34 |
| OV.47 | 10.2602111304454602,EA-J86-21105D,1600 | BAHRUL ULUM | 93.50 | 107.93 | 100.00 | 100.91 |
| OV.17 | 10.2602111146314600,EA-J86 20224-V10,2000 | EKO SUTRISNO | 89.20 | 85.77 | 100.00 | 76.51 |
| LINE.2.A | 0.OLE-Loss Time | RATNO | 9.73 | 76.24 | 100.00 | 7.42 |
| PK.01.F2 | 10.2601150021242269,F2-PK-P55-3004,2000 | RENDY T | 23.74 | 135.08 | 100.00 | 32.07 |
| P3.AF | Select WCO | BAHRUL ULUM AF | 1.76 | 6.00 | 100.00 | 0.11 |
| PL.01 | 10.2602111312124603,PL01-J86-10702V6,750 | M ALI M | 75.19 | 77.79 | 100.00 | 58.49 |
| PL.11 | 10.2602111121204596,PL11-J86-25349,3148 | EKO F | 74.27 | 54.46 | 100.00 | 40.44 |
| PL.10 | 10.2602110955164586,PL10-J86-12190 A,1800 | ANGGA W | 57.90 | 113.73 | 100.00 | 65.85 |

OEE Monitoring Dashboard

The OEE monitoring system was developed by integrating shopfloor-level PLCs with a centralized MES-oriented data architecture. PLCs were configured to capture machine status signals, production counts, cycle time, and downtime conditions. Data was transmitted from the PLCs to a background application developed in C# using a socket-based communication protocol. The application processed machine logic, executed stored procedures, and managed data transactions within SQL Server to ensure real-time and accurate data recording. The stored data was then utilized for dynamic dashboard visualization, allowing management to monitor production performance, downtime history, cycle efficiency, and machine speed across 52 production machines. This architecture ensured seamless integration from shopfloor control systems to enterprise-level performance monitoring.




Architecture OEE Dashboard

# Certification and Training

Kementerian Perindustrian REPUBLIK INDONESIA | METI | AOTS | PIDI 4.0

SERTIFIKAT

NOMOR: 653/BPSDMI.2/LEMMI-BATCH 1/IX/2023

diberikan kepada:

Raihan Anwar

Sebagai Peserta

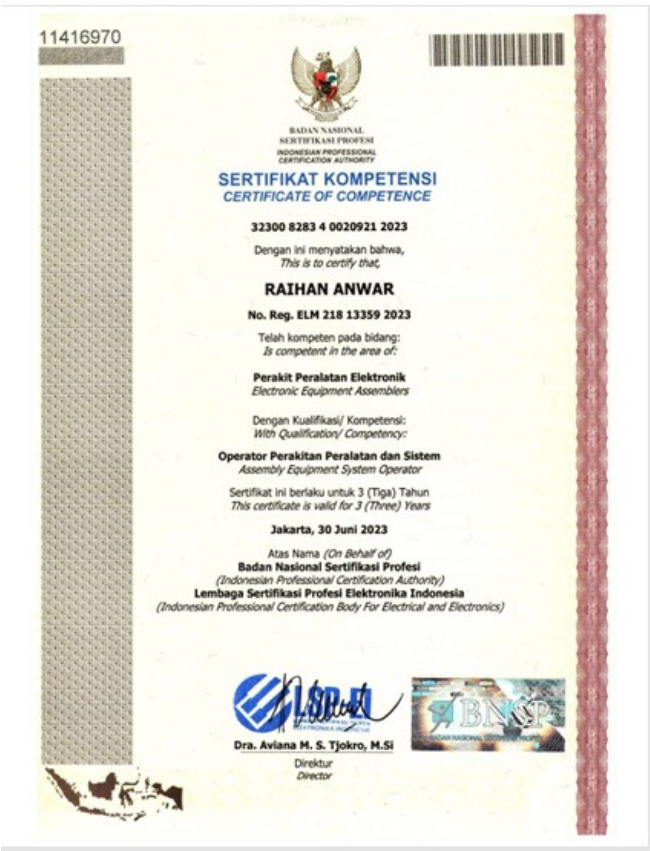
Pelatihan New Lean Manufacturing (LeMMI 4.0)
Yang diselenggarakan oleh Pusat Industri Digital Indonesia (PIDI) 4.0 Kementerian Perindustrian
Senin-Jumat/ 4 - 8 September 2023 di Gedung PIDI 4.0

KEPALA
PUSAT PENDIDIKAN DAN PELATIHAN SDM INDUSTRI
Arnes Lukman
NIP. 197503252005021001

SENIOR MANAGING DIRECTOR
THE ASSOCIATION FOR OVERSEAS TECHNICAL COOPERATION AND SUSTAINABLE PARTNERSHIPS [AOTS]
Tateishi Joji

Scanned with CamScanner

**Lean Manufacturing LeMMI 4.0 – Kemenperin**



11416970

BADAN NASIONAL SERTIFIKASI PROFESI
INDONESIAN PROFESSIONAL CERTIFICATION AUTHORITY

SERTIFIKAT KOMPETENSI
CERTIFICATE OF COMPETENCE

32300 8283 4 0020921 2023

Dengan ini menyatakan bahwa,
*This is to certify that,*

RAIHAN ANWAR

No. Reg. ELM 218 13359 2023

Telah kompeten pada bidang:
*Is competent in the area of:*

**Perakit Peralatan Elektronik**
*Electronic Equipment Assemblers*

Dengan Kualifikasi/ Kompetensi:
*With Qualification/ Competency:*

**Operator Perakitan Peralatan dan Sistem**
*Assembly Equipment System Operator*

Sertifikat ini berlaku untuk 3 (Tiga) Tahun
*This certificate is valid for 3 (Three) Years*

Jakarta, 30 Juni 2023

Atas Nama *(On Behalf of)*
**Badan Nasional Sertifikasi Profesi**
*(Indonesian Professional Certification Authority)*
**Lembaga Sertifikasi Profesi Elektronika Indonesia**
*(Indonesian Professional Certification Body For Electrical and Electronics)*

Dra. Aviana M. S. Tjokro, M.Si
Direktur
*Director*

**Daftar Unit Kompetensi*)**
*List of Units of Competency*

| NO. | Kode Unit<br>*Unit Code* | Judul Unit<br>*Unit Title* |
|---|---|---|
| 1. | C.282900.012.01 | Memelihara Lingkungan Tempat Kerja<br>*Maintaining the Workplace Environment* |
| 2. | C.282900.013.01 | Memelihara Efektivitas Hubungan di Tempat Kerja<br>*Maintaining the Effectiveness of Relationships at Workplace* |
| 3. | C.282900.028.01 | Merakit Peralatan dan Sistem Kelistrikan<br>*Assembling Electrical Equipment and Systems* |
| 4. | C.282900.029.01 | Merakit Peralatan dan Sistem Pneumatik<br>*Assembling Pneumatic Tools and Systems* |
| 5. | C.282900.030.01 | Merakit Peralatan dan Sistem Elektronik<br>*Assembling Electronic Equipment and Systems* |
| 6. | C.282900.031.01 | Merakit Peralatan dan Sistem Hidrolik<br>*Assembling Hydraulic Equipment and Systems* |

*) SKKNI Kategori Industri Pengolahan Golongan Pokok Industri Mesin Dan Perlengkapan Yang Tidak Diklasifikasikan Di Tempat Lain (YTDL) Bidang Otomasi Industri (Nomor 631 Tahun 2016)

Jakarta, 30 Juni 2023

**Lembaga Sertifikasi Profesi Elektronika Indonesia**
*Indonesian Professional Certification Body For Electrical and Electronics*

RAIHAN ANWAR
Tanda tangan pemilik
*(Signature of holder)*

Muhammad Satiri
Manajer Sertifikasi
*(Certification Manager)*

**Hydraulic & Pneumatic System Certification (BNSP)**

12309727

BADAN NASIONAL SERTIFIKASI PROFESI
INDONESIAN PROFESSIONAL CERTIFICATION AUTHORITY

SERTIFIKAT KOMPETENSI
CERTIFICATE OF COMPETENCE

J.61200 2153 7 0001296 2024

Dengan ini menyatakan bahwa,
*This is to certify that,*

**Raihan Anwar**
**No.Reg.TEL 1596 00054 2024**

Telah kompeten pada bidang:
*Is competent in the area of:*

**Teknologi Informasi dan Komunikasi**
**Information and Communication Technology**

Dengan Kualifikasi / Kompetensi :
*With Qualification / Competency of:*

**Insinyur Komputasi Awan**
**Cloud Computing Engineer**

Sertifikat ini berlaku: 3 (tiga) Tahun
*This certificate is valid for: 3 (three) Years*

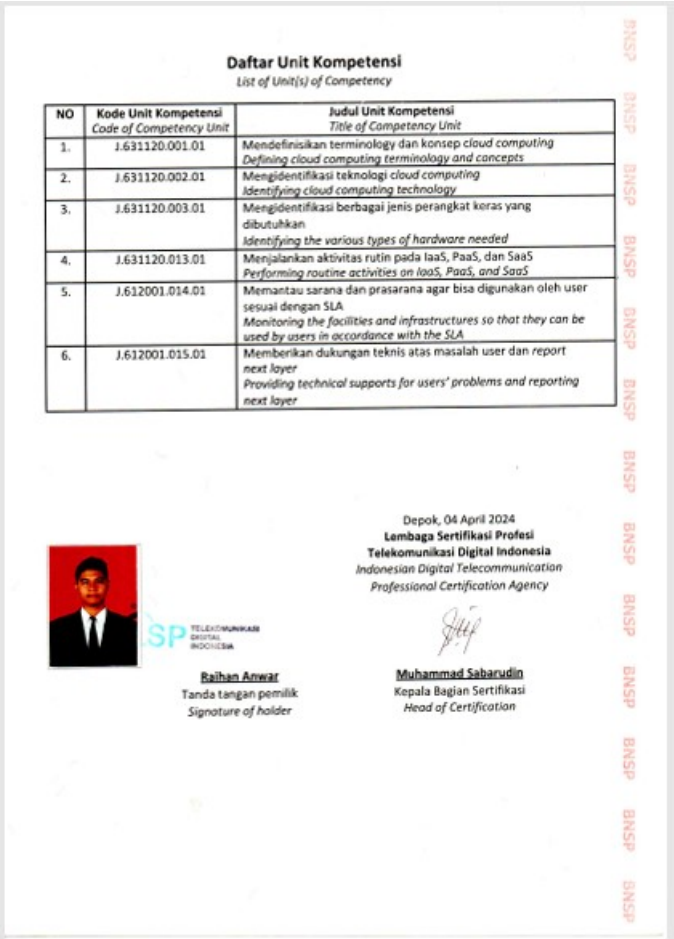
Depok, 04 April 2024

**Atas nama Badan Nasional Sertifikasi Profesi**
*On Behalf of the National Professional Certification Authority*
**Lembaga Sertifikasi Profesi Telekomunikasi Digital Indonesia**
*Indonesian Digital Telecommunication Professional Certification Agency*

LSP TELEKOMUNIKASI DIGITAL INDONESIA

BNSP

**Lingga Wardhana**
Ketua
Chairman



**Daftar Unit Kompetensi**
*List of Unit(s) of Competency*

| NO | Kode Unit Kompetensi<br>*Code of Competency Unit* | Judul Unit Kompetensi<br>*Title of Competency Unit* |
|---|---|---|
| 1. | J.631120.001.01 | Mendefinisikan terminology dan konsep cloud computing<br>*Defining cloud computing terminology and concepts* |
| 2. | J.631120.002.01 | Mengidentifikasi teknologi cloud computing<br>*Identifying cloud computing technology* |
| 3. | J.631120.003.01 | Mengidentifikasi berbagai jenis perangkat keras yang dibutuhkan<br>*Identifying the various types of hardware needed* |
| 4. | J.631120.013.01 | Menjalankan aktivitas rutin pada IaaS, PaaS, dan SaaS<br>*Performing routine activities on IaaS, PaaS, and SaaS* |
| 5. | J.612001.014.01 | Memantau sarana dan prasarana agar bisa digunakan oleh user sesuai dengan SLA<br>*Monitoring the facilities and infrastructures so that they can be used by users in accordance with the SLA* |
| 6. | J.612001.015.01 | Memberikan dukungan teknis atas masalah user dan report next layer<br>*Providing technical supports for users' problems and reporting next layer* |

Depok, 04 April 2024
**Lembaga Sertifikasi Profesi**
**Telekomunikasi Digital Indonesia**
*Indonesian Digital Telecommunication*
*Professional Certification Agency*

LSP TELEKOMUNIKASI DIGITAL INDONESIA

**Raihan Anwar**
Tanda tangan pemilik
*Signature of holder*

**Muhammad Sabarudin**
Kepala Bagian Sertifikasi
*Head of Certification*

**Cloud Computing Engineer Certification (BNSP)**

14600241

BADAN NASIONAL SERTIFIKASI PROFESI
INDONESIAN PROFESSIONAL CERTIFICATION AUTHORITY

SERTIFIKAT KOMPETENSI
CERTIFICATE OF COMPETENCE

63111 2521 6 0002890 2025

Dengan ini menyatakan bahwa,
*This is to certify that,*

**Raihan Anwar**
**No.Reg.JTEL 1596 00570 2025**

Telah kompeten pada bidang:
*Is competent in the area of:*

**Teknologi Informasi dan Komunikasi**
***Information and Communication Technology***

Dengan Kualifikasi / Kompetensi :
*With Qualification / Competency of:*

**Ilmuwan Data**
**Data Scientist**

Sertifikat ini berlaku: 3 (tiga) Tahun
*This certificate is valid for: 3 (three) Years*

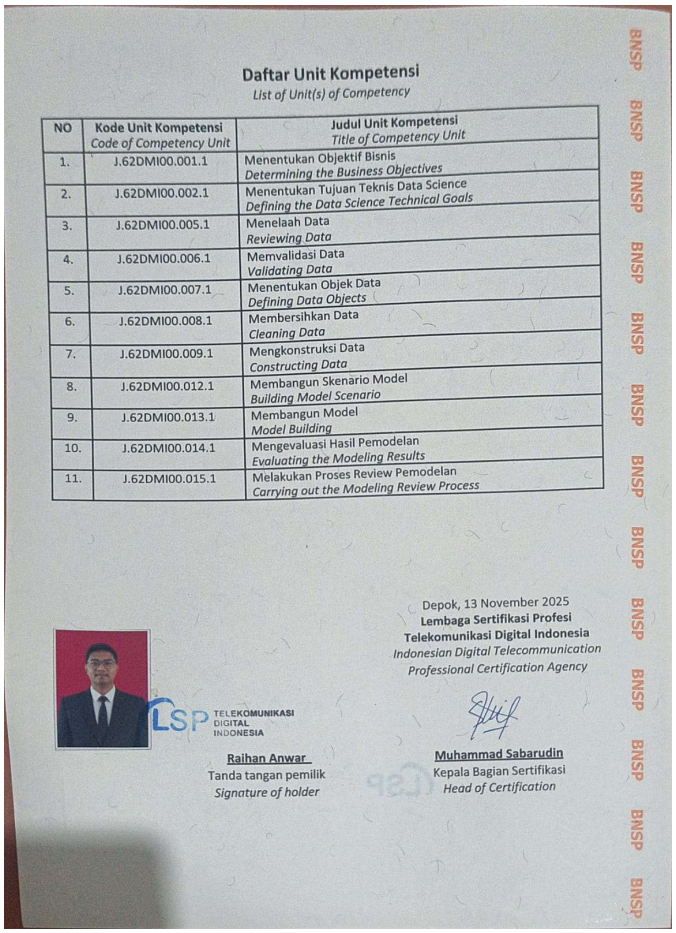
Depok,13 November 2025

**Atas nama Badan Nasional Sertifikasi Profesi**
*On Behalf of the National Professional Certification Authority*
**Lembaga Sertifikasi Profesi Telekomunikasi Digital Indonesia**
*Indonesian Digital Telecommunication Professional Certification Agency*

LSP TELEKOMUNIKASI DIGITAL INDONESIA

BNSP

**Lingga Wardhana**
Ketua
Chairman



**Daftar Unit Kompetensi**
*List of Unit(s) of Competency*

| NO | Kode Unit Kompetensi<br>*Code of Competency Unit* | Judul Unit Kompetensi<br>*Title of Competency Unit* |
|---|---|---|
| 1. | J.62DMI00.001.1 | Menentukan Objektif Bisnis<br>*Determining the Business Objectives* |
| 2. | J.62DMI00.002.1 | Menentukan Tujuan Teknis Data Science<br>*Defining the Data Science Technical Goals* |
| 3. | J.62DMI00.005.1 | Menelaah Data<br>*Reviewing Data* |
| 4. | J.62DMI00.006.1 | Memvalidasi Data<br>*Validating Data* |
| 5. | J.62DMI00.007.1 | Menentukan Objek Data<br>*Defining Data Objects* |
| 6. | J.62DMI00.008.1 | Membersihkan Data<br>*Cleaning Data* |
| 7. | J.62DMI00.009.1 | Mengkonstruksi Data<br>*Constructing Data* |
| 8. | J.62DMI00.012.1 | Membangun Skenario Model<br>*Building Model Scenario* |
| 9. | J.62DMI00.013.1 | Membangun Model<br>*Model Building* |
| 10. | J.62DMI00.014.1 | Mengevaluasi Hasil Pemodelan<br>*Evaluating the Modeling Results* |
| 11. | J.62DMI00.015.1 | Melakukan Proses Review Pemodelan<br>*Carrying out the Modeling Review Process* |

Depok, 13 November 2025
**Lembaga Sertifikasi Profesi**
**Telekomunikasi Digital Indonesia**
*Indonesian Digital Telecommunication*
*Professional Certification Agency*

LSP TELEKOMUNIKASI DIGITAL INDONESIA

**Raihan Anwar**
Tanda tangan pemilik
*Signature of holder*

**Muhammad Sabarudin**
Kepala Bagian Sertifikasi
*Head of Certification*

**Data Scientist Certification (BNSP)**